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25 April 1980
Superseding
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1 December 1961

MILITARY SPECIFICATION

RACK, BOMB EJECTOR, AIRCRAFT; AERO 7A AND AERO 7B SERIES

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification establishes the requirements for the manufacture and inspection of two types of bomb racks capable of carrying and releasing stores having suspension lugs spaced 355.6 millimeters (14 inches) and 762 millimeters (30 inches) apart and weighing from 113 kilograms (250 pounds) to 1,810 kilograms (4,000 pounds). The test limits in this specification define minimum acceptable capabilities.

1.2 Classification. Bomb ejector racks, hereinafter called bomb racks, covered by this specification shall be of the following types:

Type I AERO 7A Bomb Ejector Rack

Type II AERO 7B Bomb Ejector Rack

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The Following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ENGINEERING SPECIFICATIONS AND STANDARDS DEPARTMENT (CODE 93) NAVAL AIR ENGINEERING CENTER, LAKEHURST, N.J. 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1095

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SPECIFICATIONS

Military

MIL-T-7743	Testing, Store Suspension and Release Equipment; General Specification for
MIL-C-81842	Connector Assemblies for Bomb Rack Electric Fuzing Provisions
MIL-H-85042	Hooks, Bomb Rack, General Specification for

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MS3314	Lug, Suspension, 450 kg. (1,000 lb) class, Airborne Equipment

DRAWINGS

Naval Air Systems Command

1380540	Lug, Suspension, Mk 3 Mod 0

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Order of precedence of specifications and standards. Should a conflict exist between any requirement in this specification and any requirement in the applicable documents referenced herein, the requirements of this specification shall take precedence over the referenced (applicable) document.

3. REQUIREMENTS

3.1 First article. The bomb racks furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.3).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Interchangeability. All parts having the same part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.4 Identification of parts. All parts shall be marked as specified on the drawings listed in the data list for the bomb rack being procured, or in accordance with MIL-STD-130, if not specified on the drawings. The item identification and part number requirements of MIL-STD-100 shall govern the part number and changes thereto.

3.5 Bomb rack.

3.5.1 Construction. The construction of the bomb rack shall be in accordance with the drawings listed in the data list for the bomb rack being procured and as specified herein (see 6.2.1g).

3.5.2 Performance. The bomb racks shall satisfy all performance requirements when subjected to the following inspections in the order specified in Tables I and II.

a. Bomb Rack Inspection (see 4.6.1)

b. Functional Test (See 4.6.2)

c. Arming Mechanism (see 4.6.3)

d. Electric Fuze Arming (see 4.6.4)

e. Manual Function (see 4.6.5a)

f. Electrical Characteristics (see 4.6.5b)

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g. Lag Time or Time of Release Determination (see 4.6.5c)

h. High-g Release (see 4.6.5d)

i. Icing (see 4.6.5e)

j. Sand (see 4.6.5f)

k. Salt Spray (see 4.6.5g)

l. Shock (see 4.6.5h)

m. Vibration (see 4.6.5i)

n. Life (see 4.6.5j)

o. Static (see 4.6.5k)

3.6 Bomb rack hooks. The construction and inspection of the bomb rack hooks shall be in accordance with MIL-H-85042.

3.7 Electric fuze arming unit. The construction and inspection of the electric fuze arming unit shall be in accordance with MIL-C-81842.

3.8 Workmanship. The workmanship displayed in fabrication and assembly of the bomb racks shall be such as to assure, within design limitations, the ability of the bomb racks to meet their performance requirements under all applicable environmental conditions specified herein. Unauthorized repair, welding, heavy burrs, or parts assemblied by introduction of high stresses not prescribed in the drawings, are typical signs of inferior workmanship and shall be cause for rejection. The standards of workmanship exhibited in the approved first article sample, subject to any qualification stated in the government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship. (See 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection of the bomb racks shall be classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall consist of all of the inspections listed in Table I. First article inspection report format shall be in accordance with MIL-STD-831 (see 6.3).

4.3.1 Sampling for first article inspection. First article bomb rack samples shall consist of the first two bomb racks manufactured on contract. The bomb racks shall be permanently labeled A and B for ease of identification. Failure of the bomb racks to pass any of the first article inspections shall be cause for rejection (see 6.2.2.1).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following inspections:

a. Individual Inspections (see 4.4.1)

b. Sampling Inspections (see 4.4.2)

Quality conformance inspection report format shall be in accordance with MIL-STD-831(see 6.3).

4.4.1 Individual inspections. Individual inspections are those inspections conducted on each bomb rack. Individual inspections for the bomb racks are shown in Table II, Group 1. Failure to pass any of these inspections shall be cause for rejection of the bomb rack.

4.4.2 Sampling inspections. A random sampling of the bomb racks shall be selected by the procuring activity from each production lot in accordance with Table III and shall be subjected to the inspections listed in Table II, Group 1 and Group 2. Group 1 inspections shall be conducted prior to Group 2 inspections. Failure to pass any of these inspections shall be cause for rejection of the entire lot (see 6.2.1d).

4.5 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Room temperature. Room ambient of 25±10°C (77±18°F) indicated.

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b. Test temperature. All parts of the test item shall be stabilized at the specified temperature ±2.8°C (±5°F) indicated prior to conducting any tests. Unless otherwise specified, temperature stabilization will have been attained when the indicated temperature of the surface of the largest mass of the test item does not change by more than ±2.8°C (±5°F) in a period of 1 hour.

c. Swaybraces. Whenever tests require that a store be mounted to the bomb rack, swaybraces shall be used. The swaybraces shall be adjusted as described in 4.6.2d.

d. Test stores. The stores, as referenced herein by weight, shall conform to the following parameters. The store suspension lugs shall be in accordance with MS3314 and Drawing 1380540.

1. The 113 kilogram (250 pound) store shall utilize applicable suspension lugs, spaced 355.6 millimeters (14 inches) apart. Its center of gravity shall be based on the MK81 L.D.G.P. bomb.

2. The 450 kilogram (1,000 pound) store shall utilize applicable suspension lugs, spaced 355.6 millimeters (14 inches) apart. Its center of gravity shall be based on the MK83 L.D.G.P. bomb.

3. The 910 kilogram (2,000 pound) store shall utilize applicable suspension lugs, spaced 762 millimeters (30 inches) apart. Its center of gravity shall be based on the MK84 L.D.G.P. bomb.

4. The 1,620 kilogram (3,575 pound) store shall utilize applicable suspension lugs spaced 762 millimeters (30 inches) apart. Its center of gravity shall be midway between the suspension lugs and 203 millimeters (8 inches) below the hook bearing surface of each suspension lug.

e. Input power requirements. The bomb rack shall perform satisfactorily under all applicable conditions specified herein from a 28 VDC rated voltage power source with characteristics and limits as defined in MIL-STD-704, except that the bomb rack shall operate over a range of 20 to 30 VDC, measured at the bomb rack's input connector(s).

4.6 Inspection

4.6.1 Bomb rack inspection. Each bomb rack shall be inspected to ensure that it has been properly assembled and adjusted, that the electrical connections are secure and that the workmanship is as specified herein (see 3.8).

4.6.1.1 Qualification by analysis, inspection, and demonstration. All requirements for the bomb rack not qualified by testing shall be qualified by analysis, inspection, or demonstrations approved by the procuring activity.

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4.6.1.2 Packaging inspection. Each bomb rack shall be examined to ensure that the preparation for delivery conforms to this specification (see Section 5).

4.6.2 Functional test. Each bomb rack, mounted in the horizontal position, shall pass the following tests.

WARNING

The bomb rack shall not be fired without a reactive store load imposed on its ejector foot.

a. The hooks shall be mechanically latched and the safety pin installed. The sear latch lines shall be visually inspected to verify that each hook is securely latched. Apply 11.3 newton meters (100 inch pounds) maximum torque to the manual release. The release mechanism shall not actuate.

b. Remove the safety pin and actuate the manual release mechanism. The release mechanism shall operate smoothly and the hooks shall open.

c. Repeat a. and b. four times.

d. Suspend a 113 kilogram (250 pound) store from 355.6 millimeter (14 inch) hooks of the bomb rack. Tighten the forward and aft sets of sway braces as follows:

1. Hand tighten all braces until the pads lightly contact the store.

2. Tighten the forward starboard pad 1/4 turn.

3. Tighten the aft port pad 1/4 turn.

4. Tighten the forward port pad 1/4 turn.

5. Tighten the aft starboard pad 1/4 turn.

6. Repeat tightening in order of 2. through 5. to attain 3/8 turn total on each swaybrace.

e. Repeat b. The store shall release.

f. Repeat d. and e. with a 910 kilogram (2,000 pound) store suspended from the 762 millimeter (30 inch) hooks.

g. With the 355.6 millimeter (14 inch) hooks latched, apply a 66,720 Newton (15,00 pound) vertical down static load. Concentrate the load midway between the hooks. The hooks shall hold the load.

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h. Repeat g. with a 88,960 Newton (20,000 pound) load on the 762 millimeter (30 inch) hooks.

i. Insert the safety pin in the bomb rack. Load the ejector assembly with two applicable impulse cartridges (see 6.2.11). Suspend a 113 kilogram (250 pound) store from the bomb rack. Swaybrace in accordance with d. above. Apply 28 VDC to the bomb rack. The impulse cartridges shall not ignite, and the store shall not be ejected.

j. Remove the safety pin. Apply 28 VDC to the bomb rack. The impulse cartridges shall ignite and the store shall be ejected. Remove, disassemble, clean and inspect the ejector assembly and surrounding areas of the bomb rack as indicated in the applicable manuals (see 6.2.1h). There shall be no damage to the bomb rack.

k. Repeat i. and j. with a 910 kilogram (2,000 pound) store suspended from the 762 millimeter (30 inch) hooks.

4.6.3 Arming mechanism. A MK-9 or MK-11 bomb arming wire assembly shall be inserted in the tail and nose arming units of the bomb rack (see 6.2.1m). With each arming unit unenergized, pull each arming wire slowly, in a vertically downward direction, parallel to the side of the bomb rack. Each arming unit shall support a 40 Newton (9 pound) load, but release the wire at any load in excess of 58 Newtons (13 pounds). A minimum of three successive measurements shall be obtained on each unenergized arming unit. With each arming unit energized, pull on each arming wire in the same direction as specified above. Each arming unit shall support a load of 667 Newtons (150 pounds).

4.6.4 Electric fuze arming. The 0° pull-out force for the electric fuze connector assembly and the pull test fixture shall be in accordance with MIL-C-81842. Three successive measurements shall be made. In addition, the interlock switches (both sets of hooks) shall be verified to be open when the bomb rack is latched and closed when the bomb rack is released.

4.6.5. Additional tests. The bomb racks shall pass the following tests of MIL-T-7743 as specified, except as modified herein. The arming units shall meet the requirements of 4.6.3 when they are required to be tested, and they shall be tested in the vertically downward direction only. When required, the minimum voltage shall be 20 VDC. Applicable impulse cartridges shall be used when required (see 6.2.11).

a. Manual Function (see 4.6.5.1)

b. Electrical Characteristics (see 4.6.5.2)

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c. Lag Time or Time of Release Determination (see 4.6.5.3)

d. High-g Release (see 4.6.5.4)

e. Icing (see 4.6.5.5)

f. Sand

g. Salt Spray

h. Shock (see 4.6.5.6)

i. Vibration (see 4.6.5.7)

j. Life (see 4.6.5.8)

k. Static (see 4.6.5.9)

4.6.5.1 Manual function. This test shall be conducted in accordance with the ground service operation portion of the manual function test.

4.6.5.2 Electrical characteristics. The minimum operating voltage shall be determined by demonstrating that all electrical equipment on the bomb rack operates when 20 VDC is applied to the terminals of the applicable bomb rack connectors.

4.6.5.3 Lag time or time of release determination. This test shall be conducted in accordance with the lag or time of release determination test for dive bomb (DB) items. The 113 kilogram (250 pound) and 450 kilogram (1,000 pound) store shall be used for 355.6 millimeter (14 inch) suspension tests. The 910 kilogram (2,000 pound) and 1,620 kilogram (3,575 pound) store shall be used for 762 millimeter (30 inch) suspension tests.

4.6.5.4 High-g release. This test shall be conducted with the following conditions:

a. 355.6 millimeter (14 inch) suspension - Perform 12 ejections with the applied load based on a 660 kilogram (1,450 pound) store.

b. 762 millimeter (30 inch) suspension - Perform 13 ejections with the applied load based on a 1,620 kilogram (3,575 pound) store.

4.6.5.5 Icing. This test shall be conducted in accordance with the Navy and Air Force icing test.

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4.6.5.6 Shock. Light load tests shall be conducted with a 113 kilogram (250 pound) store for the 355.6 millimeter (14 inch) suspension hooks. Yield load tests shall be conducted with a 1,620 kilogram (3,575 pound) store for the 762 millimeter (30 inch) suspension hooks. Applied shock amplitudes shall be as shown in Table. IV.

4.6.5.7 Vibration. This test shall be conducted in accordance with Procedure II of the vibration test. The minimum weight store used shall be 113 kilograms (250 pounds) for the 355.6 millimeter (14 inch) hooks and the maximum weight store used shall be 450 kilograms (1,000 pounds) for the 762 millimeter (30 inch) hooks.

4.6.5.8 Life test. This test shall be conducted in accordance with the life test for explosive items. A 113 kilogram (250 pound) store, for the 355.6 millimeter (14 inch) hooks, shall be used for light store ejections. A 1,620 kilogram (3,575 pound) store, for the 762 millimeter (30 inch) hooks, shall be used for heavy store ejections.

4.6.5.9 Static test. This test shall be conducted with the following conditions. Yield loads shall be 1.15 times and ultimate loads 1.5 times the limit loads given in Table V. The combined loads and moments shall be applied to a rigid simulated store of negligible weight (91 kilograms, 200 pounds maximum) in accordance with Table V such that they act at the store center of gravity.

4.6.6 Qualification cross reference. Table VI identifies each requirement in Section 3 and correlates the Section 4 paragraph that qualify it.

5. PACKAGING

5.1 Bomb rack preservation and packaging. Preservation and packaging shall be in accordance with Level A, B or C as specified in MIL-STD-794, as specified in the contract, without the use of preservation compound.

5.2 Packing. Shipping containers shall contain one each packaged bomb rack. The container shall be suitable for Level A, B or C as specified in MIL-STD-794.

5.3 Unitized loading. Unitized loading for the bomb racks is authorized as specified in MIL-STD-794.

5.4 Marking. All shipping containers shall be marked in accordance with the requirements of MIL-STD-129 with the following information:

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RACK, BOMB, EJECTOR, AIRCRAFT AERO
MILITARY SPECIFICATION NO. MIL-R-22622(AS)
PRESERVED (DATE)
DOMESTIC OR EXPORT PACKED (AS APPLICABLE)
GOVERNMENT ORDER NO. (OR CONTRACT NO. IF ORDER NO. IS NOT ASSIGNED)
NAME OF SUPPLIER (AND NAME OF MANUFACTURER IF NOT THE SAME)
MANUFACTURER'S CODE IDENTIFICATION NUMBER

6. NOTES

6.1 Intended use. The bomb racks covered by this specification are intended for use in all types of bombing. They are mounted externally in an aircraft wing pylon or fuselage. They are intended for use on either land-based or carrier-based aircraft.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

c. Samples subjected to sampling inspection shall not be considered or accepted as part of the contract (see 4.4.2)

d. Sampling inspection selection (see 4.4.2)

e. Number of samples and inspection sequence for production lots in excess of 300 units (see Table III).

f. Inspection sequence for hooks. Prior to installation into bomb racks, the hooks shall have met all of the first article and sampling requirements of MIL-H-85042.

g. Applicable drawing package for type of bomb rack required (see 3.5.1).

h. Applicable maintenance manuals for type of bomb rack required (see 4.6.2).

i. Name and location of government representative responsible for random selection of inspection samples (see 4.4.2).

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j. Items of data required (see 6.3).

k. Name and location of government approved test laboratory.

l. Number and designation of impulse cartridges required for testing (see 4.6.2 and 4.6.5).

m. Name and location of government activity responsible for supplying arming wire assemblies (see 4.6.3).

6.2.2 Contract provision. Contracts shall specify the following provision for first article inspection.

6.2.2.1 First article. When a first article is required for inspection and approval (see 3.1, 4.3, 6.2 and 6.3), the contract shall specify the following provision for first article inspection. When a contractor is in continuous production of bomb racks from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate:

a. If first article inspections are conducted at the contractor's plant or a government approved laboratory, an inspection report shall be forwarded to the procuring activity for verification.

b. That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirments identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirement List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.3	First Article Inspection Reports	DI-T-5329- Inspection Test Reports
4.4	Quality Conformance Inspection Reports	DI-T-5329-Inspection Test Reports

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(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.4 Quality assurance definitions. Definitions for quality assurance terms are in accordance with MIL-STD-109.

6.5 International standardization agreement. Certain provisions (3.5.1) of this specification are the subject of international standardization agreements STANAG 3575, AIR STD 20/10 and AIR STD 20/17. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Preparing Activity
NAVY-AS
DOD Proj. No. 1095-N077

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TABLE I

FIRST ARTICLE INSPECTIONS - SCHEDULE AND SEQUENCE 1/

TESTS	TEST PARAGRAPH	SAMPLE	
		A	B
Bomb Rack Inspection	4.6.1	1	1
Functional	4.6.2	2	2
Arming Mechanism	4.6.3	3	3
Electric Fuze Arming	4.6.4	4	4
Manual Function	4.6.5a	5	X
Electrical Characteristics	4.6.5b	X	5
Lag Time or Time of Release Determination	4.6.5c	6	X
High-g Release	4.6.5d	X	6
Icing	4.6.5e	7	X
Sand	4.6.5f	8	X
Salt Spray	4.6.5g	9	X
Shock	4.6.5h	X	7
Vibration	4.6.5i	X	8
Life	4.6.5j	10	X
Static	4.6.5k	X	9

1/ Inspection sequence is denoted by the numbers in the sample columns. An X denotes inspection not required.

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TABLE II

QUALITY CONFORMANCE INSPECTIONS - SCHEDULE AND SEQUENCE 1/

TESTS	TEST PARGRAPH	SAMPLE			
GROUP 1		All Samples			
Bomb Rack Inspection	4.6.1	1			
Functional	4.6.2	2			
Arming Mechanism	4.6.3	3			
Electric Fuze Arming	4.6.4	4			
GROUP 2		A	B	C	D
Manual Function	4.6.5a	1	X	1	X
Electrical Characteristics	4.6.5b	X	1	X	1
Lag Time or Time of Release Determination	4.6.5c	2	X	X	X
High-g Release	4.6.5d	X	2	X	2
Icing	4.6.5e	3	X	X	X
Sand	4.6.5f	4	X	X	X
Salt Spray	4.6.5g	5	X	2	X
Shock	4.6.5h	X	3	X	3
Vibration	4.6.5i	X	4	X	X
Life	4.6.5j	6	X	3	4
Static	4.6.5k	X	5	X	5

1/ Inspection sequence is denoted by the numbers in the sample columns. An X denotes inspection not required.

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TABLE III

SAMPLING TEST ITEMS

ITEM	PRODUCTION LOT SIZE	NUMBER OF SAMPLES
Bomb Racks	100 or less 101 - 200 201 - 300 301 or More	2 3 4 1/

1/ The procuring activity shall specify the number of samples, and the sequence of inspections each unit shall be subjected to for lots in excess of 300 units (see 6.2.1e).

TABLE IV

APPLIED SHOCK LOADS

DIRECTION	APPLIED SHOCK LIGHT LOAD (g's) FOR 113 KG. (250 LB.) STORE	APPLIED SHOCK YIELD LOAD (g's) FOR 1,620 KG. (3,575 LB.) STORE
Vertical Down	25	5.8
Vertical UP	25	2.3
Longitudinal Forward	25	6.3
Longitudinal Aft	25	6.3
Transverse Port	11.6	1.2
Transverse Starboard	11.6	1.2

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TABLE V

STATIC TEST LIMIT LOADS

SIMULATED STORE CHARACTERISTICS [1]

Store	Lug Centers	Store Diameter	Vertical Dimension, Lug Bearing Surface to Store C.G.
A	355.6 mm (14 in.)	775 mm (30.5 in.)	413 mm [1] (16.25 in.) [1]
B	762.0 mm (30 in.)	355.6 mm (14.0 in.)	203 mm [1] (8.0 in.) [1]
C	762.0 mm (30 in.)	- -	MER, with six MK82 Stores [2]

TYPE I BOMB RACK
COMBINED LIMIT LOADS AND MOMENTS ABOUT REFERENCE POINT (STORE C.G.)

Condition	Store	P_z Vertical	P_y Side	P_x Longitudinal	M_{zz} Yaw Moment	M_{yy} Pitch Moment	M_{xx} Roll Moment
1	A	-38,542 N (-8,665 lbs.)	-20,229 N (-4,548 lbs.)	7,232 N (1,626 lbs.)	-336 N·m (-2,974 inch-lbs.)	-2,864 N·m (-25,348 inch-lbs.)	- -
2	B	-115,804 N (-26,035 lbs.)	-5,413 N (-1,217 lbs.)	15,221 N (3,422 lbs.)	-2,821 N·m (-24,965 inch-lbs.)	-2,183 N·m (-19,322 inch-lbs.)	-
3	A	-8,896 N (-2,000 lbs.)	- -	51,481 N (11,574 lbs.)	- -	- -	- -
4	B	-18,722 N (-4,209 lbs.)	- -	108,069 N (24,296 lbs.)	- -	- -	- -
5	A	-16,382 N (-3,683 lbs.)	-8,433 N (-1,896 lbs.)	-46,802 N (-10,522 lbs.)	1,965 N·m (17,391 inch-lbs.)	3,902 N·m (34,530 inch-lbs.)	-
6	B	-34,348 N (-7,722 lbs.)	-17,676 N (-3,974 lbs.)	-98,821 N (-22,217 lbs.)	2,270 N·m (20,087 inch-lbs.)	4,507 N·m (39,887 inch-lbs.)	-

TYPE II BOMB RACK
COMBINED LIMIT LOADS AND MOMENTS ABOUT REFERENCE POINT (STORE C.G.)

1	C	- -	33,649 N (7,565 lbs.)	- -	3,223 N·m (28,522 lbs.)	-14,050 N·m (-124,348 inch-lbs.)	301 N·m (2,661 inch-lbs.)
2	C	-20,345 N (-4,574 lbs.)	- -	101,726 N (22,870 lbs.)	- -	- -	- -

SIGN CONVENTION

$+P_x$ = Longitudinal Aft
$+P_y$ = Side to Port
$+P_z$ = Vertical Up
$+M_{xx}$ = Roll Clockwise Looking Forward
$+M_{yy}$ = Pitch Nose Up
$+M_{zz}$ = Yaw Nose to Port

[1] The store C.G. shall be on the longitudinal centerline of the store midway between the suspension lugs.

[2] The load is based on a MER with six MK82 stores. The centroid of this combination is 411 millimeters (16.2 inches) aft of the forward hook and 343 millimeters (13.5 inches) from the load bearing surface.

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TABLE VI

Qualification Cross Reference

Paragraph Title	Requirement Paragraph	Test Paragraph
First Article	3.1	4.3
Selection of Specs & Stds	3.2	4.6.1.1
Interchangeability	3.3	4.6.1.1
Identification of Parts	3.4	4.6.1.1
Construction	3.5.1	4.6.1.1
Bomb Rack Inspection	3.5.2a	4.6.1
Functional Test	3.5.2b	4.6.2
Arming Mechanism	3.5.2c	4.6.3
Electric Fuze Arming	3.5.2d	4.6.4
Manual Function	3.5.2e	4.6.5a
Electrical Characteristics	3.5.2f	4.6.5b
Lag Time or Time of Release Determination	3.5.2g	4.6.5c
High-g release	3.5.2h	4.6.5d
Icing	3.5.2i	4.6.5e
Sand	3.5.2j	4.6.5f
Salt Spray	3.5.2k	4.6.5g
Shock	3.5.2l	4.6.5h
Vibration	3.5.2m	4.6.5i
Life	3.5.2n	4.6.5j
Static	3.5.2o	4.6.5k
Bomb Rack Hooks	3.6	4.6.1.1
Electric Fuze Arming Unit	3.7	4.6.1.1
Workmanship	3.8	4.6.1

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DOCUMENT IDENTIFIER AND TITLE

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

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